DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CareerPilot

AI-Assisted Career Guidance, Job Matching, Skill-Gap Analysis and Application Support

Project Team

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CP

RESUME

JOBS

SKILLS

1 PROBLEM STATEMENT

Problem Definition

- Job seekers must manually move between resumes, job portals, skill lists and learning resources.
- Matching a resume to a job is difficult when requirements are spread across different descriptions.
- Skill gaps are often identified manually, without a personalized learning path.
- Application drafts can introduce claims that are not supported by the candidate's resume.
- Career workflows lack one continuous path from profile → matching → learning → application review.

Business / Real-World Context

- Students and fresh graduates searching for relevant entry-level roles.
- Candidates who need guidance on missing skills and interview preparation.
- Career services that need structured, repeatable candidate support.
- Applicants who want tailored resumes, cover letters and application answers.
- Human review remains important before an application is treated as submitted.

USER AND STAKEHOLDER ANALYSIS

Target Users

- Students / fresh graduates
- Job seekers changing roles
- Candidates preparing for interviews
- Applicants needing application support
- Career mentors / placement teams

Stakeholder Identification

Stakeholder	Main Need
Candidate	Find relevant jobs, understand gaps, prepare applications
Career Mentor	Review recommendations and learning direction
Placement Team	Support repeatable candidate workflows
Recruitment Context	Improve relevance of candidate/job matching
Human Reviewer	Approve application materials before submission

3 PROPOSED SYSTEM: ANALYSIS PIPELINE

CareerPilot transforms a PDF resume into a guided career workflow: profile → job matching → skill gaps → learning → reviewed application materials.

1 Resume Input

Upload PDF and extract candidate profile.

2 ProfileAgent

Parse skills, education and experience into structured data.

3 JobAgent

Search listings and rank jobs using the ML matcher.

4 LearningAgent

Compare job requirements with candidate skills.

5 RAG + Groq

Retrieve relevant career-guide sections and explain the gap.

6 Application + Review

Generate materials, verify claims and wait for approval.

4 PROPOSED SYSTEM: DATA SOURCES

The prototype combines local project data, career guides, uploaded resumes and optional live job providers.

Upload Resume

Extracts education, skills, projects and experience from the candidate's PDF. The resume provides evidence for checking LLM-generated claims.

Local Job Dataset

jobs.csv: Contains 800 synthetic job listings with job titles, required skills, locations and experience requirements. Supports local job matching and ranking.

Live Job APIs

Retrieves current vacancies from Remotive and Adzuna, including job descriptions and application links.

Career Knowledge Base

Seven Markdown guides cover Python, data analysis, machine learning, interviews, Linux security, data engineering, cloud and React. Agentic RAG retrieves relevant sections when the LLM decides

SQLite Database:

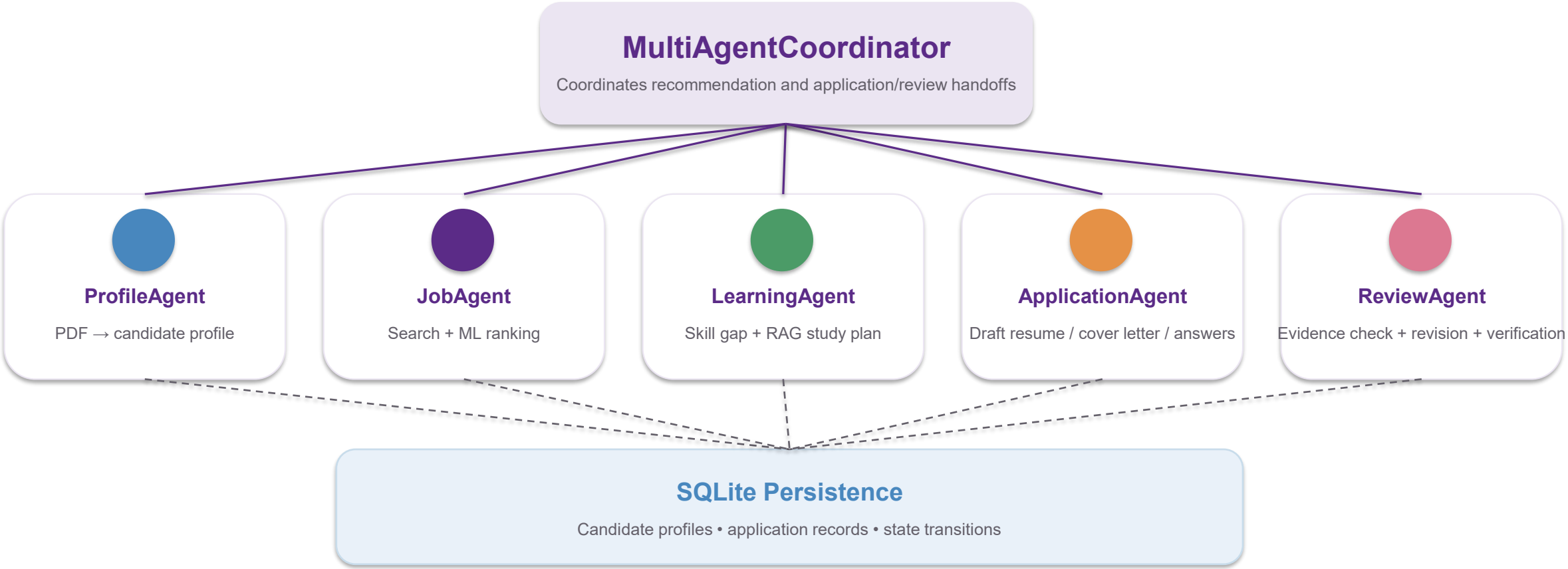
Stores candidate profiles, saved jobs, generated applications, review evidence, draft revisions, human approval and submission status.

5 TOOLS AND TECHNOLOGIES USED

1	**User Layer**	HTML/CSS/JavaScript • static browser UI
2	**Application Layer**	Flask • Python • REST-style API routes
3	**Agent Layer**	ProfileAgent • JobAgent • LearningAgent • ApplicationAgent • ReviewAgent
4	**ML Layer**	Random Forest JobMatcher • rule-based skill comparison
5	**Knowledge Layer**	Markdown KnowledgeBase • lexical TF/IDF-style retrieval • Groq
6	**Data / Integration**	SQLite • CSV • Remotive • Adzuna • PDF extraction

AGENT ARCHITECTURE

Specialized agents divide the career workflow into focused responsibilities while the coordinator manages handoffs.



AGENT RESPONSIBILITIES

ProfileAgent

Extracts a structured candidate profile from an uploaded PDF.

JobAgent

Searches available listings and asks the matcher to rank them.

LearningAgent

Computes skill gaps, decides whether retrieval helps and generates a study plan.

ApplicationAgent

Creates tailored resume, cover letter and application answers.

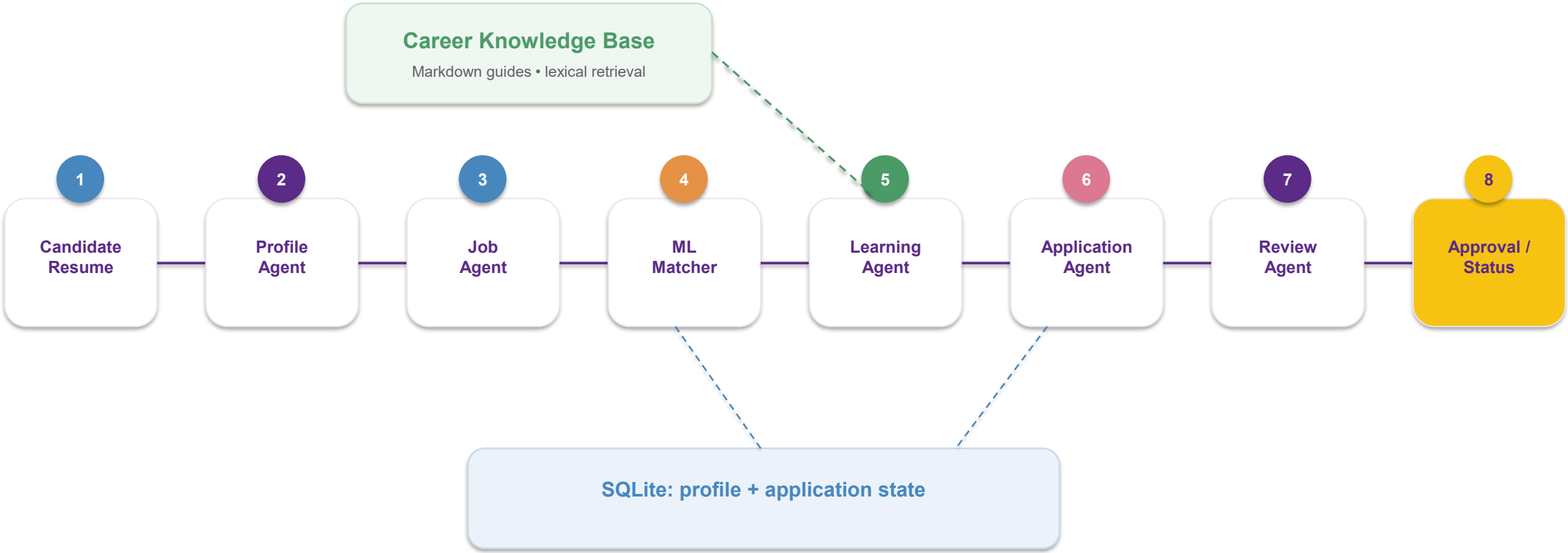
ReviewAgent

Checks generated text against resume evidence and requests revisions.

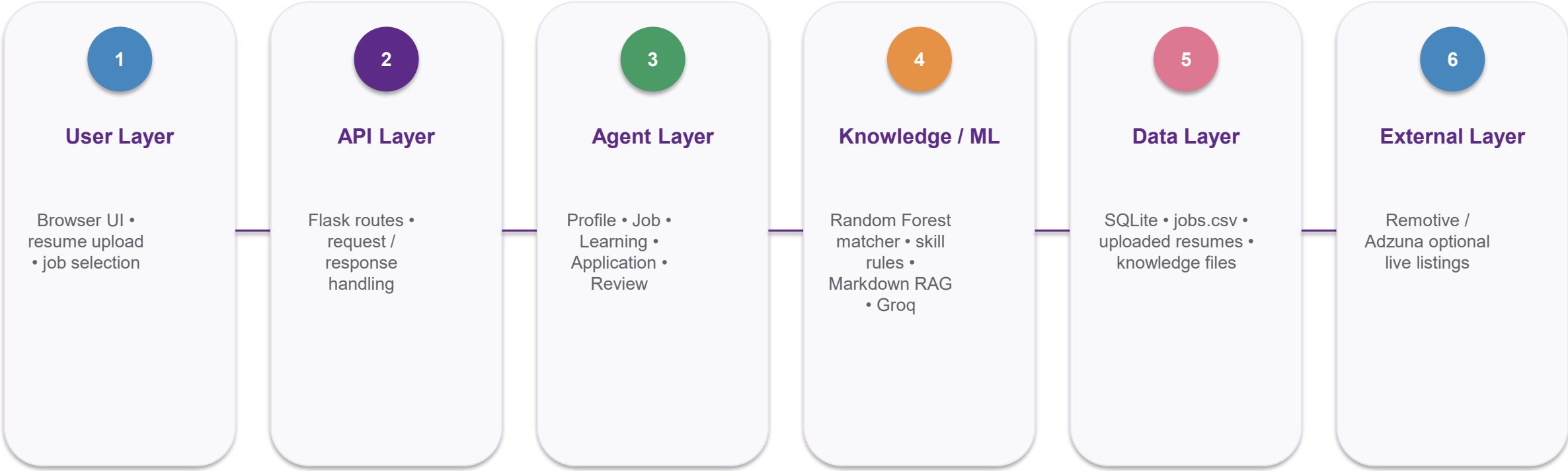
MultiAgentCoordinator

Coordinates recommendation and application/review workflows.

8 AGENT WORKFLOW AND ORCHESTRATION



9 SYSTEM ARCHITECTURE: TIERED VIEW



AGENTIC RAG IMPLEMENTATION

1 Knowledge Sources

Knowledge Base: Loads local career guides from knowledge base and splits them into sections using Markdown headings.
Gap Analysis: Compares the candidate's profile with the selected job to identify missing skills and experience gaps.

2 Conditional Retrieval

LLM Retrieval Decision: The Learning Agent asks the LLM whether additional guidance is needed. It returns a retrieval decision, focused query and reason.
Conditional Retrieval: When requested, lexical search ranks guide sections by query-term relevance and returns up to three matches.

3 Explanation + Study Plan

Personalized Learning Plan: The LLM uses the identified gaps and retrieved excerpts to generate an explanation and ordered study plan. Without excerpts, it provides general guidance.
Source Transparency: The interface displays retrieval decisions and matched sections. Source names are retained only when they belong to retrieved documents.

11 SYSTEM IMPLEMENTATION

Main Technologies

Flask | Python | Random Forest | SQLite | Groq | Markdown RAG

Implementation Flow



Functional Prototype

1. **Resume upload**
   PDF-only upload, max 5 MB
2. **Job recommendations**
   Local or optional live listings
3. **Skill-gap analysis**
   Missing skills + learning plan
4. **Application drafting**
   Resume, cover letter, answers
5. **Review + approval**
   Evidence check before submitted state

COMPARISON & INNOVATION

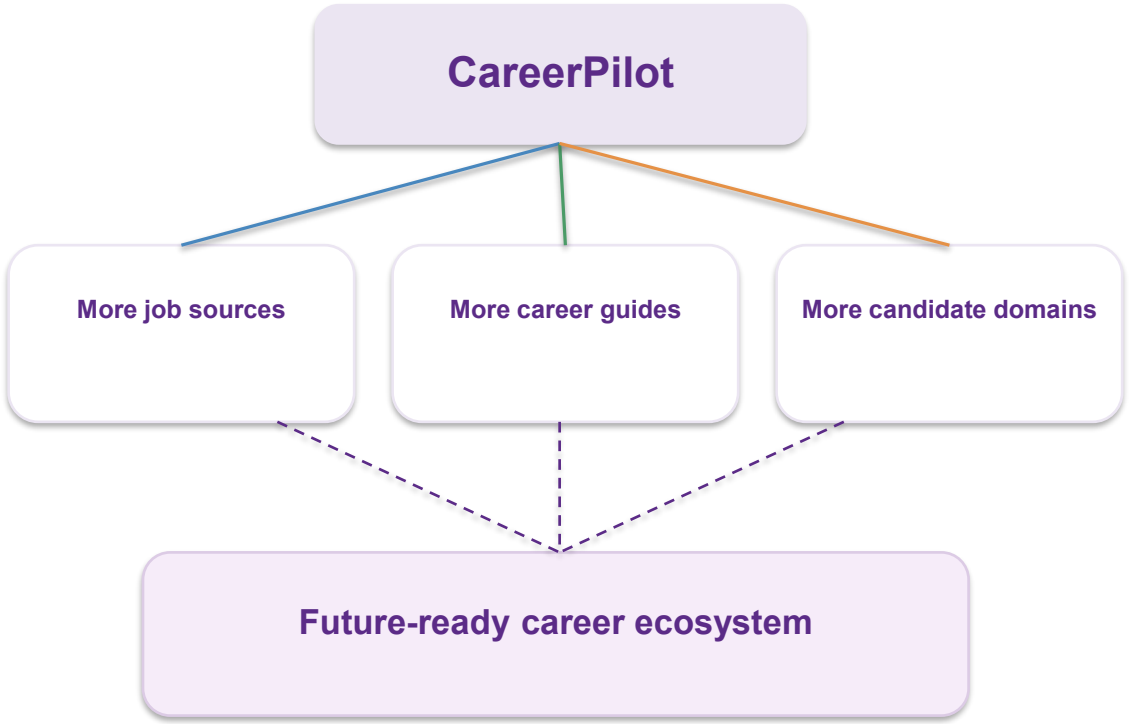
Feature	Manual Workflow	CareerPilot
Resume understanding	Manual	✓ PDF parsing
Job retrieval	Separate portals	✓ Local + optional live
ML ranking	✗	✓ Random Forest
Skill-gap analysis	Manual	✓ Job-specific
Career-guide retrieval	Separate	✓ Conditional local RAG
Application drafting	Manual	✓ Tailored
Evidence review	Manual	✓ ReviewAgent
Human approval	✓	✓ Required before submitted

SCALABILITY & REAL-WORLD

Real-World Applicability

- Student placement and campus recruitment preparation
- Personalized job discovery for graduates
- Skill-gap based upskilling plans
- Interview preparation using local career guides
- Application drafting with evidence review

Scalability



14 FUTURE WORK

1. Personalized Job Alerts

Notify users about new openings based on their skills, preferred role, location and salary expectations.

2. AI Interview Preparation

Generate role-specific questions, conduct mock interviews and provide feedback.

3. Learning Progress Dashboard

Track completed courses and projects, then update skill gaps and job recommendations.

4. Application Management

Add deadline reminders, interview schedules and follow-up tracking.

5. Recruiter Connection

Let employers post verified vacancies and connect with suitable candidates, with job seekers controlling profile visibility.

CONCLUSION

CareerPilot demonstrates how AI can move beyond isolated chatbot responses toward a structured career-support workflow.

The system:

- parses a PDF resume into a structured candidate profile
- searches and ranks jobs using a Random Forest matcher
- identifies job-specific skill gaps
- conditionally retrieves local career-guide content
- generates a personalized learning plan
- drafts tailored application materials
- reviews generated claims against resume evidence
- requires explicit candidate approval before submitted status

CareerPilot

Thank You

CareerPilot — AI-Assisted Career Guidance

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